Predictive Analysis of Vehicle Insurance Demand Using Machine Learning Techniques

Mingwei Sun
Xi'an Jiaotong-Liverpool University
Suzhou, Jiangsu, China
Mingwei.Sun2102@student.xjtlu.edu.cn

Abstract

Health insurance and vehicle insurance play a crucial role in reducing economic burdens, alleviating psychological stress, and maintaining social stability. Given the similarities in buyer characteristics for both types of insurance, predicting whether customers who purchase health insurance are interested in vehicle insurance is of significant importance for companies offering both products. By analyzing data from the Kaggle website, this paper employs machine learning techniques, including XGBoost, AdaBoost, and Multi-Layer Perceptron (MLP), to predict customer interest in purchasing vehicle insurance and compares the performance of different models. The study results indicate that AdaBoost performs best in predicting vehicle insurance demand, followed by XGBoost, while MLP performs relatively weaker in this task. These findings provide important insights for insurance companies to optimize marketing strategies, adjust pricing models, and better manage risks. However, future research should focus on integrating more advanced technologies and improving data quality to further enhance the performance of predictive models.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches.

Keywords

Health Insurance, Vehicle Insurance, Machine learning, Purchase Perception, Data Analyze

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1 Introduction

Health insurance, also known as medical insurance, is designed to cover medical expenses resulting from illness or accidental injury. It provides financial support to the insured, reducing their burden by covering costs such as doctor visits, hospitalizations, prescription drugs, and surgeries. The importance of health insurance is embodied in its ability to alleviate economic pressure, encourage preventive care, mitigate psychological stress, contribute to social stability, and promote healthier lifestyles [1]. At the same time, another type of insurance also focuses on the financial loss of the insured. The vehicle insurance is a kind of property insurance, the main purpose of which is to provide financial compensation to the owner of the insured vehicle when the vehicle suffers damage due to an agreed insured incident such as natural disasters or accidental casualties. Various types of insurance such as vehicle damage insurance, third party liability insurance, theft insurance, vehicle occupants' liability insurance, etc. usually includes in it. Vehicle insurance's importance is mainly reflected in the fact that it can alleviate the financial losses of vehicle owners after an accident and provide effective financial protection for the insured. Secondly, it can transfer many of the risks may face in the use of vehicles to the insurance company. In addition to this, some types of insurance, such as third-party liability insurance, can enable third parties to be compensated for the losses caused by the insured vehicle and protect the interests of the third party, thus maintaining the stability of the society. Both types of insurance are purchased by people who want to reduce their possible financial risk, and have similarities in terms of purchaser characteristics. In these cases, predicting whether customers who purchase health insurance are also interested in vehicle insurance is crucial for insurance companies who provides both insurances.

Machine learning plays a pivotal role in these predictive tasks. Machine learning algorithms can recognize patterns and trends that may not be immediately observed by human analysts. These insights enable companies to predict with high accuracy which customers are likely to purchase vehicle insurance based on their health insurance profiles. The application of machine learning techniques allows for continuous learning and adaptation, improving the prediction models over time as more data becomes available.

The methods of machine learning, such as decision trees, random forests, and neural networks can process complex datasets in order to uncover hidden correlations between different types of insurance purchases [2]. This capability is invaluable for insurance companies aiming to personalize their marketing strategies and tailor their product offerings to individual customer needs. Furthermore, machine learning models can segment customers based on their likelihood to purchase vehicle insurance, allowing for targeted marketing efforts that maximize conversion rates and customer satisfaction. The implications of these predictive analyses for insurance companies are profound. From a business strategy perspective, accurate predictions enable companies to allocate resources more

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efficiently, investing in areas with the highest potential returns. Companies can develop and refine products that align closely with customer interests, enhancing the overall value proposition. Additionally, targeted marketing campaigns based on machine learning predictions can reduce customer acquisition costs and improve the return on investment for marketing expenditures.

This paper will explore the predictive analysis of vehicle insurance demand using machine learning techniques. We will employ various machine learning algorithms to tackle the problem of predicting customer vehicle insurance purchases and conduct a detailed comparison of the performance of different machine learning models.

2 Previous Works

There are various methods employed in predicting insurance purchase intentions, each with distinct advantages and limitations. Commonly used techniques include linear regression, logistic regression, decision trees, and support vector machine (SVM), which have been extensively explored in existing research.

Logistic regressions effective for binary classification problems, making it suitable for predicting whether an individual will purchase insurance [3] . Its simplicity and interpretability are key strengths, though it assumes a linear relationship between variables, which may not always be accurate. Linear regression, used for predicting continuous outcomes, is also straightforward and interpretable but sensitive to outliers and assumes linearity. The researcher used linear regression to inspect the effect of different factors on the purchase of life insurance and takaful annuities by academic staff of public universities in Malaysia and used these methods to fit the predictions. The results indicated that the motivation to save for the purchase of insurance has a stronger relationship with religiosity, while the R-square is 0.780. And these indicate that the independent variable of the study explains 78 per cent of the variance of the dependent variable, which is a high degree of goodness of fit [4].

Decision trees are intuitive and easy to interpret, handling both numerical and categorical data with minimal preprocessing. Random forests, combined with multiple decision trees, improve their accuracy and robustness by averaging results, though they can be computationally intensive and less interpretable[5] . In the study of predicting individuals who will purchase earthquake insurance and the related factors that influence earthquake insurance purchases. The researcher used a decision tree and random forest-based approach and the two models showed better prediction of earthquake insurance purchases with accuracies of 89.55 percent and 100 percent [6].

SVM and multilayer perceptron are also powerful model. SVM are effective in high-dimensional spaces and useful for classification problems with clear margin separation, handling non-linear relationships through kernel functions[7] . Mare, C. et al. proposed an approach based on support vector machines as well as multilayer perceptron classifiers to predict whether Romanian farmers will purchase crop insurance. And the prediction is based on crop-related and farmer-related variables [8]. These two models achieved the highest accuracy of all the models applied, 0.74 and 0.76, respectively. Multilayer Perceptron, a type of neural network, excel at capturing non-linear relationships and modeling complex patterns [9]. They require significant computational resources and large amounts of data, so that can be challenging for them to interpret and tune properly.

These studies emphasize the efficacy of machine learning and statistical models in predicting insurance demand. They highlight the importance of selecting appropriate techniques under specific features of the data and the context of the insurance product being analyzed. As we proceed with our analysis of vehicle insurance demand, we will build on these methodologies to develop models that can accurately predict purchase intentions and identify the key factors driving these decisions.

3 Dataset

The data used in this study is publicly available (Health Insurance Cross Sell Prediction, https://www.kaggle.com/datasets/anmolkumar/health-insurance-cross-sell-prediction). The data is based on customers of an insurance company that offers health insurance, and we will use this dataset to predict whether or not customers would be interested in the company's auto insurance.

The dataset consists of 381109 samples and 11 variables that are divided into three groups, the first being demographically related variables such as gender, age, and area of living; the second being vehicle related variables such as whether or not they have a driver license and a car, the age of the car, and whether or not the car has been damaged; and the last being company policy related variables such as the annual premium the customer pays, the channels through which they purchase their insurance, and whether they are a part of the company's policy.

We performed preprocessing on these data. The categorical data are converted to numeric data. For example, gender is converted to 1 and 0. For vehicle age, less than one year is set to 1, one to two years is set to 2, and more than two years is set to 3 respectively. Then, the data will be divided into a training set and a test set. The test set will contain 20% of the original data and the rest is the training set.

Among all the variables, the variable that shows the interest of purchasing vehicle insurance is Response. The value of it equals to one represents that the customer is interested in the insurance, and equal to zero if the client is not interested. Of the other variables, the one most closely related to Response is whether or not the customer has experienced a vehicle damage. As shown in Figure 1, the samples with no desire to purchase vehicle insurance are significantly more likely not experienced a vehicle damage, while the vast majority of the sample interested in vehicle insurance had experienced. This also suggests that car accidents are an important reason for customers to consider purchasing vehicle insurance.

4 Methods

In this study, we employed three advanced machine learning techniques to predict vehicle insurance demand: XGBoost, AdaBoost, and Multi-Layer Perceptron (MLP). The unique advantages of each method help to improve the accuracy and reliability of predictive models.

Firstly, XGBoost is a scalable and efficient implementation of gradient boosting machines [10]. Its popularity stems from its efficiency

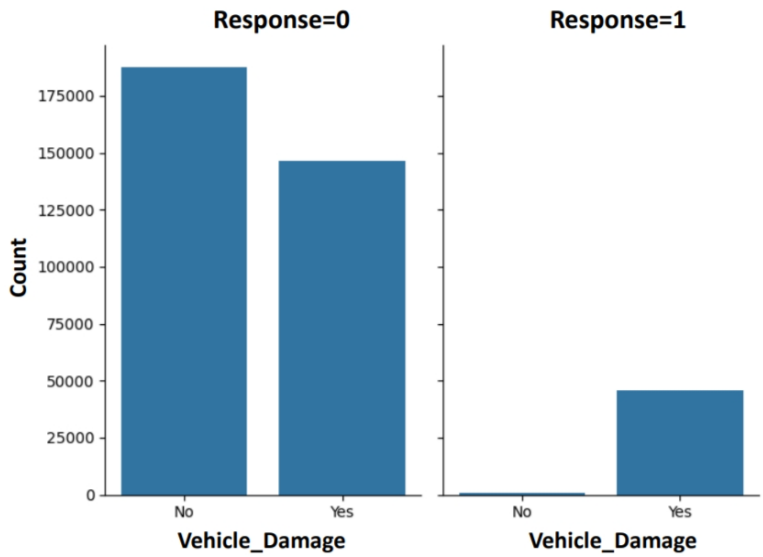

Figure 1: The illustration of response and vehicle damage.

and high performance, particularly in various machine learning tasks involving structured data. The primary advantages of XGBoost include its processing speed and performance, attributed to its parallel processing capability, which significantly reduces training time. Additionally, this algorithm can naturally handle missing values, making it robust in real-world datasets where missing data is common. The regularization of L1 and L2 in XGBoost helps to prevent overfitting and improves its generalization capabilities. Moreover, its flexibility allows it to be used for both classification and regression problems, effectively handling large-scale datasets. In the meantime, pruning and regular term is added to efficiently prevent overfitting. The objective function of XGBoost is mainly composed of loss function L and regular term Ω, where n is the number of samples and t is the number of the decision trees.

$$Obj = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i\right) + \sum\nolimits_{i=1}^{t} \Omega\left(f_i\right) \tag{1}$$

It mainly uses the value of the tth and $t-1th$ prediction to make a prediction for the sample and the formula can be shown as [10]:

$$\hat{y}_i^t = \hat{y}_i^{t-1} + f_t\left(x_i\right) \tag{2}$$

Since it is feature-based, the features are sorted before each computation, thus speeding up the follow-up.

Secondly, AdaBoost is a boosting algorithm that enhances robustness and accuracy by combining the predictions of multiple base learners [11]. The principle of AdaBoost involves initialize the weights first and then adjust the sample weight distributions $W_{t+1}(x)$ according to the error rate e_t of the previous round:

$$W_i = \frac{1}{N} \tag{3}$$

$$e_t = \frac{\sum_{i=1}^{N} W_i \times I\left(y_i \neq c_t\left(x_i\right)\right)}{\sum_{i=1}^{N} W_i} \quad x = 1, 2, 3, \ldots, N \tag{4}$$

$$w_t = \log\left(\frac{1-e_t}{e_t}\right) \tag{5}$$

$$W_{t+1}\left(x\right) = W_t\left(x\right) \times \exp\left(w_t \times I\left(y_i \neq c_t\left(x_i\right)\right)\right) \;\; x = 1, 2, 3, \ldots, N \tag{6}$$

where W_i is the weight distribution of the previous round, $I()$ is the indicator function get 1 when conditions are met and 0 in all other cases, e_t is the error rate of the weak classifier $c_t(x_i)$ and w_t is its weight. Then sequentially combing all trained weak classifiers to form a strong classifier $C(x)$, with each classifier focusing more on the instances that were misclassified by previous classifiers.

$$C\left(x\right) = \arg\max_{k} \sum_{t=1}^{M} w_t \times I\left(y_i = k\right) \tag{7}$$

The simplicity of AdaBoost makes it relatively easy to implement and understand, making it a good starting point for boosting algorithms. Additionally, the algorithm adapts to the data by focusing more on difficult cases, improving overall predictive performance. By combining weak learners, AdaBoost can create a strong predictive model even from simple base classifiers, and then significantly improve the classification accuracy. The versatility of this algorithm allows it to be used with various types of base learners, enhancing its flexibility.

Lastly, MLP is a type of feedforward artificial neural network composed of multiple layers of neurons, which includes an input layer, one or more hidden layers, and an output layer [9]. In one layer, each neuron is correspondently connected with the neurons in the subsequent layer. MLPs typically use backpropagation algorithm combined with gradient descent to optimize model parameters. During the training process, predictions are calculated by forward propagation and then backpropagated based on the errors. This process is repeated several times until a predetermined number of training rounds or some stopping condition is reached, the default value for the number is usually 100. The main advantage of MLP is that it can model complex non-linear relationships through

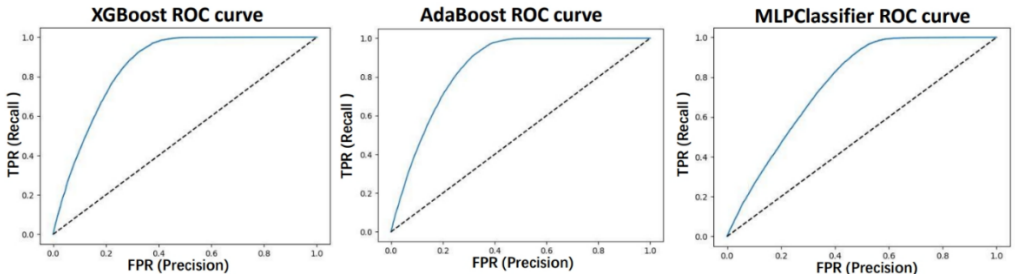

Figure 2: The results of 3 different method.

its multi-layer structure and non-linear activation functions. The high flexibility and universality of MLP enables its application to a wide range of problem domains, including regression, classification, and time series prediction. MLPs can learn hierarchical representations, capturing intricate patterns in the data. With sufficient data and computational resources, MLPs can be scaled to handle very large and complex datasets.

5 Results

The evaluation of the model performance was according to the accuracy score and the Area Under the Receiver Operating Characteristic Curve (AUC). AUC is illustrated in Figure 2.

XGBoost demonstrated a strong performance with an accuracy score of 87.64% and an AUC of 0.8557. This method's efficiency, handling of missing values, and regularization capabilities contribute to its robust performance. The ROC curve for XGBoost shows a high true positive rate across a wide range of false positive rates, indicating good discriminatory ability.

AdaBoost achieved an accuracy score of 87.70% and an AUC of 0.8533. AdaBoost's adaptability and simplicity in combining weak learners to create a strong predictive model make it a competitive choice. The ROC curve for AdaBoost also indicates strong performance, similar to XGBoost, with a slightly lower AUC.

MLP yielded an accuracy score of 87.70% and an AUC of 0.7674. While MLP showed comparable accuracy to XGBoost and AdaBoost, its AUC was lower, suggesting that it has less discriminatory power in this context. The primary strengths of MLP did not translate into superior performance in this specific task.

Based on the results, AdaBoost emerges as the optimal method for predicting vehicle insurance demand. Although its accuracy is marginally higher than that of XGBoost, the difference is not substantial. Both methods performed well, but AdaBoost's simplicity and slightly better accuracy make it the preferred choice. However, it is worth noting that XGBoost also remains a strong contender due to its high AUC and robust handling of data irregularities. MLP, despite its capabilities in modeling non-linear relationships, did not perform as well in this scenario, as evidenced by its lower AUC. As one of the most basic neural networks, MLP has some limitations as the problems become more complexed and the amount of data increases. The first is its high computational complexity. As the number of neural network layers and parameters rises, there are more computational difficulties, leading to slower training and prediction. The second is the risk of overfitting. If the amount of data is insufficient or lack of regularization methods, MLP has a high risk of overfitting, which makes its predictive ability decrease. In addition, tuning the parameters is also a major challenge for MLP, which are very sensitive to the setting of hyperparameters and require careful tuning and setting to achieve the best performance. At the same time, the high demand for data pre-processing makes its training effect and generalization ability easy to be limited. Whereas AdaBoost and XGBoost use an ensemble learning approach that combines multiple single integrated methods to improve the overall performance of the model. These giving them higher stability and generalization than MLP.

6 Conclusion

This study investigated the predictive performance of three machine learning techniques—XGBoost, AdaBoost, and MLP—in forecasting vehicle insurance demand. The main findings indicate that AdaBoost achieved the highest accuracy score of 87.70%, closely followed by XGBoost with 87.64%, and MLP with 87.70%. And for Area Under the Curve (AUC), XGBoost led with 0.8557, AdaBoost was slightly behind at 0.8533, and MLP trailed with 0.7674. These results demonstrate that both AdaBoost and XGBoost are highly effective for this predictive task, with AdaBoost slightly edging out in overall performance.

The potential value of these models in practical applications is significant. Accurate prediction of vehicle insurance demand can help insurance companies optimize their marketing strategies, adjust their pricing models, and better manage risk. The models, particularly AdaBoost and XGBoost, can be integrated into insurance companies when making decisions to enhance efficiency and profitability. However, there is still room for improvement. Fine-tuning hyperparameters, incorporating more features, and utilizing more sophisticated ensemble methods could further enhance model performance. For the dataset used in our study, the amount of data is large and there are many variables, some variables such as the age of the vehicle is a range of data, but it is replaced by three numbers, to some extent this will affect the accuracy of prediction study. Also, there may be many other influencing factors in predicting, also resulting in a lower level of accuracy in our prediction. For the learning prediction methods used, we chose two methods based on boosting, which makes the use of machine learning methods relatively homogeneous.

For future research, several directions and challenges are worth exploring. First, combining these models with other advanced techniques such as deep learning or hybrid models could provide deeper insights and improved accuracy. Second, addressing data

quality and ensuring the robustness of the models in different market conditions will be crucial for real-world applications. Lastly, the challenge of interpretability remains significant; developing models that are not only accurate but also transparent and interpretable will be essential for gaining trust and ensuring regulatory compliance.

In conclusion, while AdaBoost emerges as the optimal method in this study, XGBoost also proves to be a strong contender. The implementation of these models in practical settings holds considerable promise, though future research must address technical challenges and strive for continual improvement.

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